Recap: Fourier series

- If $f(x)$ is piecewise smooth on $(-L, L]$ then its Fourier series is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x), \qquad k_n = \frac{n\pi}{L}$$

  where $f(x)$ is continuous

- The coefficients are given by Euler-Fourier formulae
- We denote the periodic extension of $f(x)$ as $F(x)$
- If $F(x)$ continuous then Fourier series convergence is uniform (e.g., $f = x^2$)
- If $F(x)$ is discontinuous then Gibbs phenomenon occurs (e.g., $f = x$)

Lecture 18

In Lecture 18, we will:

1. Define complex Fourier series
2. Define even/odd periodic extensions of functions and calculate their Fourier series

Complex exponentials create a more compact Fourier series

Definition

- The function $f(x)$ on $(-L, L]$ has the *complex Fourier series*

$$\sum_{n=-\infty}^{n=\infty} c_n \mathrm{e}^{\mathrm{i}k_n x} \quad \text{where} \quad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\mathrm{e}^{-\mathrm{i}\pi k_n x}\mathrm{d}x$$

- This will be our starting point for Fourier transforms (Topic 6)

Exercise 4.6: Calculate the complex Fourier series of $f(x) = x$ on $(-\pi, \pi]$

Fourier series of even and odd functions

Consider the Fourier series of $f(x)$ on $(-L, L]$, i.e.,

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x) \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x$$

Case f odd, i.e., $f(-x) = -f(x)$

The Fourier series simplifies to

$$\sum_{n=1}^{\infty} b_n \sin(k_n x) \quad \text{where} \quad b_n = \frac{2}{L}\int_{0}^{L} f(x)\sin(k_n x)\mathrm{d}x$$

Even and odd extensions

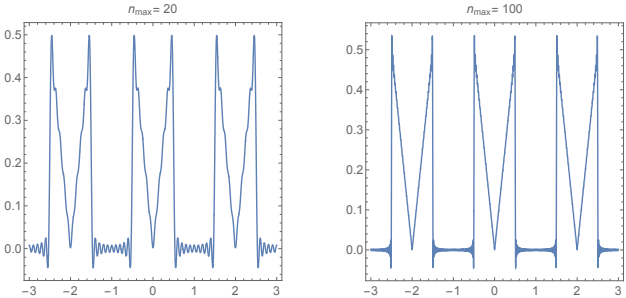
- Let $f(x)$ be defined on $x \in [0, L]$
- We can calculate its *even periodic extension* $F(x)$, such that

$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

- We can also calculate its *odd periodic extension* $F(x)$, such that

$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= -f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

Exercise 4.7: Find the Fourier series of the even periodic extension of

$$f(t) = \begin{cases} t, & 0 \leqslant t \leqslant 1/2 \\ 0, & 1/2 < t \leqslant 1 \end{cases}$$

Exercise 4.8: Evaluate the series $\displaystyle\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2}$